

# IAD-Risk: Risk-Aware Identity-Agenda Disentanglement for Scholarly Work Deduplication

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## Abstract

Scholarly work deduplication must decide whether two records describe the same work, not whether they merely share a research agenda. This distinction is fragile because scientific representation and pair-classification models can assign high semantic scores to distinct papers from the same task, benchmark, or citation community. We study this failure mode as *identity-agenda confusion* and propose IAD-Risk, a risk-aware framework that treats agenda relatedness as possible false-merge risk rather than direct identity evidence. IAD-Risk combines stratified data provenance, identity and agenda relation heads, agenda-non-identity hard negatives, and false-merge risk gating to block risky automatic pair merges before clustering. We evaluate the framework with IAD-Bench, a benchmark contract that separates DeepMatcher gold labels, SciRepEval/SciDocs-style proxy labels, and OpenAlex/OpenCitations silver hard negatives. In the Open-v2 evidence snapshot, the result rows are scope-bounded mechanism evidence rather than a same-scope comparative ranking: single-space scientific representation baselines show HNFMR 0.790–0.999 on the full pair scope, whereas transformer-backed IAD-Risk variants report same-work F1=0.980 and zero observed HNFMR on the held-out test scope. The results support a conservative pair-level conclusion: identity-agenda disentanglement can control risky automatic merges in the reported setting, while cluster-level quality claims require cluster artifacts and broader manual or source-heldout validation before broad method-ranking claims.

*Keywords:* scholarly entity matching, work deduplication, identity-agenda disentanglement, hard-negative false-merge rate, false-merge risk, provenance-aware evaluation, scientific document representation

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## 1. Introduction

Scholarly knowledge systems rely on entity matching to decide whether multiple records describe the same scientific work. Digital libraries, literature review tools, citation graphs, recommender systems, and evidence synthesis pipelines all require stable work identities because a false merge can combine evidence from different papers and distort downstream analysis. This paper focuses on scholarly work deduplication: given two scholarly records with titles, abstracts, authors, venues, identifiers, topics, and citations, the system must decide whether an automatic merge is safe. The operational target is therefore not only to find likely duplicates, but also to avoid false merges that contaminate downstream scholarly records.

The central technical challenge is that agenda evidence can resemble identity evidence. Two papers can share a task, dataset, method family, citation neighborhood, or OpenAlex topic while making different contributions. Scientific representation models such as SPECTER and SciNCL are valuable because they bring related papers close in embedding space [1, 2]; that behavior supports retrieval and candidate generation, but becomes unsafe when semantic closeness is used as merge evidence. This failure is particularly important in scholarly settings because false merges are harder to repair than missed duplicates: once distinct papers are clustered as one work, citation counts, recommendation signals, and review evidence can be contaminated. Merge safety therefore requires explicit attention to hard-negative cases where two records are related but non-identical.

We formulate this failure mode as identity-agenda confusion. Identity refers to whether two records denote the same scholarly work. Agenda refers to whether two records discuss the same research problem, benchmark, method family, or citation community. Existing entity matching methods [3–5] typically optimize pair classification, while scholarly document representation methods optimize semantic relatedness. Neither objective alone makes the desired decision boundary explicit: same-agenda non-identities should increase false-merge risk rather than merge confidence. The paper therefore treats identity-agenda confusion as a merge-safety failure in scholarly deduplication, not as an ordinary semantic-similarity error.

### 1.1. Motivating Failure Case

A simple failure case makes the distinction concrete. Consider two scholarly records that share the same task or benchmark and use similar termi-

nology, but report a different contribution, model variant, or experimental claim. A scientific representation model may assign high semantic similarity because the records are close in topic space. For work deduplication, however, the safe decision is to avoid an unsafe automatic merge: without DOI agreement, version evidence, or other same-work identity evidence, the pair should be deferred or blocked rather than merged. The point is that agenda relatedness is not identity evidence; it is a risk condition that should be handled explicitly by HNFMR-style evaluation.

Table 1: Motivating failure case for identity-agenda confusion. The example is an illustrative failure case, not a prevalence estimate.

| Observed pair signal                                          | Agenda interpretation                                             | Merge-safety interpretation                                                                    |
|---------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Same task or benchmark with overlapping technical terms.      | The records are related and may receive high semantic similarity. | Relatedness alone is insufficient for automatic same-work identity.                            |
| Different contribution, model variant, or experimental claim. | The papers can still belong to one research agenda.               | The pair requires same-work identity evidence before merging.                                  |
| No shared DOI, version marker, or clear publication lineage.  | A relatedness model may still rank the pair highly.               | Treating the pair as an automatic merge is an unsafe automatic merge and contributes to HNFMR. |

We propose IAD-Risk, a risk-aware identity-agenda disentanglement framework for scholarly work deduplication. IAD-Risk learns separate relation signals for same-work identity, same-agenda relatedness, and agenda-non-identity hard negatives, then derives a false-merge risk score from these signals. A pair is eligible for automatic merging only when identity evidence is sufficiently strong and agenda-non-identity risk remains below a merge threshold. This selective decision rule turns deduplication from a high-recall similarity problem into a selective safety problem with explicit rejection thresholds.

The paper makes three contributions. First, it defines identity-agenda confusion as a targeted failure mode in scholarly entity matching and introduces hard-negative false-merge rate as a safety metric. Second, it presents IAD-Bench, a provenance-aware benchmark contract that keeps gold, proxy, silver, and human-reviewed labels separate. Third, it evaluates IAD-Risk

within a shared Open-v2 pair schema while explicitly marking row-scope differences between full-scope baselines and held-out IAD-Risk rows; clustering and artifact-audit interfaces remain reproducible extensions rather than unsupported primary evidence. The current evidence supports targeted false-merge suppression under stratified evaluation; it does not support a broad method-ranking claim or a claim that silver labels replace human gold.

Table 2: Contribution-evidence summary. The table states what each contribution is, which evidence supports it, and which stronger claim remains outside the current manuscript.

| Contribution                                                         | Supporting evidence in this paper                                                                                                         | Claim boundary                                                                                  |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Identity-agenda confusion as a scholarly deduplication failure mode. | Problem formulation, hard-negative false-merge rate, and Open-v2 hard-negative evidence.                                                  | Identifies a concrete safety failure mode, not a prevalence estimate for all scholarly corpora. |
| IAD-Bench as a provenance-aware pair contract.                       | Gold, proxy, silver, and manual-validation layers with label source, label strength, provenance, split, and hard-negative fields.         | Defines an evaluation contract; proxy and silver labels are not human identity gold.            |
| IAD-Risk as a risk-aware merge mechanism.                            | Open-v2 evidence snapshot with single-space baseline HNFMR 0.790–0.999 and held-out IAD-Risk same-work F1=0.980 with zero observed HNFMR. | Supports targeted false-merge suppression, not a broad method-ranking claim.                    |

## 2. Related Work

Entity resolution research defines the broader pipeline in which scholarly deduplication is situated. Classical record linkage formalizes the decision of whether two records refer to the same entity [3], while modern surveys describe end-to-end entity resolution as a workflow involving blocking, matching, and clustering under scale and heterogeneity constraints [6]. System work such as Magellan further emphasizes that practical entity matching requires sampling, labeling, debugging, workflow management, and production execution rather than only a matching function [7]. IAD-Risk follows this pipeline view, but focuses on a specific safety gap in scholarly data: a pair

can be highly related in topic or citation context while still being an unsafe merge.

Neural entity matching methods provide the closest algorithmic baselines for learning pair-level match decisions. DeepER represents tuples with neural distributed representations and supports blocking over learned tuple vectors [8]. DeepMatcher explores design choices for neural entity matching models [4], and Ditto shows that transformer-based matching can benefit from pretraining, fine-tuning, and data augmentation [5]. RoBERTa-style pair encoders [9] are therefore natural baselines for this paper. The distinction is that these methods usually optimize a same-entity classification objective, whereas IAD-Risk adds explicit same-agenda and agenda-non-identity heads so that semantic relatedness can become blocking evidence rather than merge evidence.

Scientific document representation methods are strong baselines because they intentionally capture citation, topical, and semantic relatedness. SPECTER uses citation-informed supervision to learn document-level scientific representations and introduced SciDocs as a document-level evaluation benchmark [1]. SciNCL improves scientific representations with neighborhood contrastive learning [2], and SciRepEval broadens scientific representation evaluation across classification, regression, ranking, and search while releasing the SPECTER2 model family [10]. These methods are useful for candidate generation and relatedness scoring, but their optimization target is not conservative work identity. IAD-Risk therefore treats high scientific relatedness as a possible false-merge risk unless identity evidence is also strong.

Scientific language resources and open scholarly metadata make reproducible benchmark construction possible, but they do not remove the need for provenance-aware evaluation. SciBERT shows that domain-specific pre-training on scientific text improves scientific NLP tasks [11], and S2ORC provides metadata, abstracts, references, and structured full text for many open-access papers [12]. OpenAlex provides a large open index of scholarly works and concepts [13], while OpenCitations provides open citation data [14]. IAD-Bench uses these resources to construct same-work, same-agenda, and agenda-non-identity evidence, but separates gold, proxy, and silver labels so that citation or topic proximity is never presented as human identity gold.

The closest-work comparison spans End-to-end entity resolution systems, Neural entity matching, Scientific document representations, and Open schol-

arly metadata benchmarks; the complete positioning matrix is reported in the supplementary material. The main distinction is the combination of false-merge risk gates with gold, proxy, and silver strata for merge-safety framing.

This positioning narrows the novelty claim to four auditable differences. First, the paper treats identity-agenda confusion as a false-merge safety failure rather than as generic semantic relatedness. Second, it evaluates that failure with HNFMR so that agenda-level hard negatives are not hidden inside aggregate pair accuracy. Third, IAD-Bench records label strength and provenance so that gold identity labels, proxy agenda evidence, and silver stress evidence remain separate. Fourth, IAD-Risk turns agenda relatedness into risk evidence through explicit relation heads and a merge gate. IAD-Risk is not a replacement for end-to-end entity resolution workflows, and it is also not a comparative ranking over all neural matching methods; the reported comparisons test the merge-safety framing under the available Open-v2 evidence snapshot. Finally, IAD-Bench does not claim that OpenAlex/OpenCitations silver evidence is human gold, but uses those rows as provenance-marked stress evidence for agenda-level false-merge risk.

### 3. Problem Formulation

Let  $d_i$  and  $d_j$  be two scholarly records. The target identity label  $y_{ij}^I$  indicates whether the records describe the same scholarly work. The agenda label  $y_{ij}^A$  indicates whether the records discuss a related research agenda. A hard negative is a pair with high agenda evidence but negative identity evidence:

$$y_{ij}^A = 1, \quad y_{ij}^I = 0.$$

This pair should not be merged even if the documents are semantically close.

We evaluate merge safety with hard-negative false-merge rate:

$$\text{HNFMR} = \frac{\#\{(i, j) : \hat{m}_{ij} = 1, y_{ij}^A = 1, y_{ij}^I = 0\}}{\#\{(i, j) : y_{ij}^A = 1, y_{ij}^I = 0\}},$$

where  $\hat{m}_{ij}$  is the automatic merge decision. The goal is not only to maximize same-work F1, but also to control HNFMR under explicit evidence provenance.

### 3.1. Notation and Relation Semantics

Table 3 summarizes the notation used throughout the paper. The key distinction is that identity variables describe whether two records denote the same work, while agenda variables describe topical or citation-level relatedness. This distinction prevents a high relatedness score from being interpreted as direct merge evidence.

Table 3: Notation and relation semantics. The paper uses separate symbols for identity, agenda relatedness, agenda-non-identity risk, and merge decisions.

| Symbol or field     | Meaning                                             | Role in evaluation                                                         |
|---------------------|-----------------------------------------------------|----------------------------------------------------------------------------|
| $d_i, d_j$          | Two scholarly records.                              | Input pair for scoring and merge decisions.                                |
| $y_{ij}^I$          | Identity label: the records describe the same work. | Gold identity evidence when supported by the label source.                 |
| $y_{ij}^A$          | Agenda label: the records share a research agenda.  | Proxy or silver relatedness evidence, not sufficient for merging.          |
| $y_{ij}^{ANI}$      | Agenda-non-identity label.                          | Hard-negative supervision for pairs that are related but should not merge. |
| $p_{\text{work}}$   | Predicted same-work probability.                    | Positive merge evidence.                                                   |
| $p_{\text{agenda}}$ | Predicted agenda-relatedness probability.           | Relatedness evidence that can increase risk when identity is weak.         |
| $p_{\text{ani}}$    | Predicted agenda-non-identity probability.          | Direct risk evidence against automatic merging.                            |
| $p_{\text{risk}}$   | Derived false-merge risk score.                     | Gate for rejecting or deferring risky pairs.                               |
| $\hat{m}_{ij}$      | Automatic merge decision.                           | Determines FMR and HNFMR.                                                  |
| label strength      | Gold, proxy, silver, or manual validation layer.    | Controls how a row can be used in training, evaluation, and claims.        |

## 4. IAD-Bench

IAD-Bench is a benchmark contract for organizing scholarly pair evidence. It is not a claim that every label has the same evidential status. Each pair stores relation labels, expected identity labels, expected agenda labels, label source, label strength, provenance, split, and hard-negative level.

Table 4: IAD-Bench label layers. Gold, proxy, and silver labels are reported separately to avoid overstating evidence strength.

| Layer             | Source                               | Use in this paper                  |
|-------------------|--------------------------------------|------------------------------------|
| Gold              | DeepMatcher / py_entitymatching      | Same-work identity evaluation      |
| Proxy             | SciRepEval / SciDocs-style proximity | Same-agenda analysis               |
| Silver            | OpenAlex / OpenCitations             | Agenda-non-identity hard negatives |
| Manual validation | Independent human review             | External validation target         |

### 4.1. Document Schema Contract

IAD-Bench first standardizes each scholarly record before constructing pair evidence. The document contract keeps identity-bearing metadata, agenda-bearing metadata, and source provenance in one record so that downstream pair construction can be audited without redistributing raw third-party files. The required fields are grouped as follows:

- record identity: `document_id` and `source_dataset`;
- text and authorship: `title`, `abstract`, and `authors`;
- bibliographic metadata: `year`, `venue`, `doi`, `arxiv_id`, and `openalex_work_id`;
- agenda context: `topics` and `references`.

Missing values are represented by empty strings, empty arrays, or null values rather than by dropping required fields. The full document schema table is reported in the supplementary material.

### 4.2. Pair Schema Contract

IAD-Bench uses a pair schema that makes relation semantics and evidence strength auditable. The required fields are grouped as follows:



- pair identity: `pair_id`, `source_document_id`, and `target_document_id`;
- relation targets: `relation_label`, `expected_label`, and `expected_agenda_label`;
- evidence source: `label_source`, `label_strength`, and `label_provenance`;
- evaluation control: `split` and `hard_negative_level`.

The schema separates the binary same-work target from agenda relatedness, records the source and strength of each label, and keeps split and hard-negative metadata visible so that training, threshold selection, held-out reporting, and manual audit cannot be silently mixed. It also identifies agenda-level hard negatives for HNFMR. The full pair schema table is reported in the supplementary material.

The Open-v2 benchmark variant used in the core evidence combines 415 DeepMatcher gold pairs with 10,000 OpenAlex-derived silver hard negatives, producing 10,415 pair records over 1,737 documents. The DeepMatcher component provides same-work gold labels, while the OpenAlex component stresses same-agenda non-identity behavior. This design is intentionally conservative: OpenAlex and OpenCitations are used to expose false-merge risk, not to replace human gold.

Table 5: Open-v2 benchmark composition used in the current evidence snapshot. The table separates identity evidence from hard-negative stress evidence before any metric is reported.

| Evidence stratum                                 | Pair count | Evaluation role                                                                            | Claim boundary                                                                                            |
|--------------------------------------------------|------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| DeepMatcher gold identity                        | 415        | Measures same-work matching ability.                                                       | Supports identity evaluation within the imported gold benchmark pairs.                                    |
| OpenAlex and OpenCitations silver hard negatives | 10,000     | Stresses agenda-level false-merge behavior.                                                | Supports hard-negative safety analysis, not human non-identity gold.                                      |
| Open-v2 combined pair scope                      | 10,415     | Provides the shared pair contract for the reported evidence snapshot over 1,737 documents. | Supports the Open-v2 mechanism demonstration; broader source-heldout claims require additional artifacts. |

## 5. Method

### 5.1. Overview

IAD-Risk converts scholarly deduplication into selective risk-aware matching. Given a candidate pair, the model estimates identity, agenda, and agenda-non-identity signals, then derives a false-merge risk score. The final merge decision is made only when identity evidence exceeds a same-work threshold and risk evidence remains below blocking thresholds. Figure 1 summarizes the workflow. The method section follows the same execution order: it first defines identity and agenda signals, then introduces the relation heads and risk score, then specifies the training and inference trace, merge gate, algorithmic audit trail, operational boundary, and implementation details.

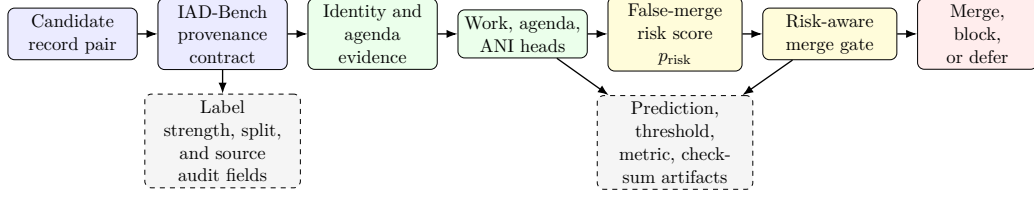


Figure 1: IAD-Risk pipeline. The framework first anchors each candidate pair in the IAD-Bench provenance contract, separates identity evidence from agenda evidence, predicts work, agenda, and ANI heads, derives a false-merge risk score, and then applies a conservative merge gate. Dashed audit nodes mark the schema fields and external artifacts required to reproduce prediction, threshold, metric, and checksum evidence.

### 5.2. Identity and Agenda Signals

IAD-Risk separates the representation space into identity-oriented and agenda-oriented evidence. Identity evidence includes strong identifiers, near-duplicate titles, author overlap, year and venue agreement, and work-level metadata. Agenda evidence includes shared research topics, similar abstracts, citation neighborhoods, shared references, and scientific representation similarity. The separation is necessary because the same feature can have different meanings: high embedding similarity may indicate same work, but it may also indicate two different papers in the same research area.

### 5.3. Multi-Task Relation Heads

The model predicts three pair-level relation quantities:  $p_{\text{work}}$ ,  $p_{\text{agenda}}$ , and  $p_{\text{ani}}$ . The same-work head estimates identity. The same-agenda head estimates topical relatedness. The agenda-non-identity head detects pairs that are related but should not be merged. The false-merge risk score is then derived from these heads:

$$p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}.$$

This construction makes the safety signal high when a pair is explicitly predicted as agenda-non-identity, or when it is agenda-related but lacks strong identity evidence.

### 5.4. Risk Score Design Rationale

$p_{\text{risk}}$  is a conservative upper-envelope risk proxy for automatic merge decisions. The score increases monotonically with agenda-non-identity evidence through  $p_{\text{ani}}$ , and it also increases when agenda evidence is high and identity

evidence is weak through  $p_{\text{agenda}}(1 - p_{\text{work}})$ . The max operator keeps either direct ANI evidence or indirect agenda-without-identity evidence sufficient to block automatic merging. This design matches the safety objective of the paper: a pair should defer rather than merge when either source of false-merge risk is high.

Table 6: Risk score design rationale. The score is designed as a conservative merge gate, not as an unconditional probability of error.

| Design element                                | Design rationale                                                                | Boundary                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| $p_{\text{ani}}$ term                         | Directly represents evidence that a related pair is not the same work.          | Requires valid ANI supervision or artifact-backed hard-negative evidence.          |
| $p_{\text{agenda}}(1 - p_{\text{work}})$ term | Flags pairs that are topically related but lack corresponding identity support. | It is not a calibrated probability unless validated against held-out artifacts.    |
| Max operator                                  | Makes either risk pathway sufficient to block a merge.                          | Conservative blocking can increase deferrals and must be tuned to the application. |
| Threshold gate                                | Converts the risk proxy into a merge, block, or defer decision.                 | Threshold transfer must be rechecked under new source distributions.               |

### 5.5. Training Objective

The learning objective combines identity, agenda, and agenda-non-identity supervision only through valid supervision channels. Let  $m_{ij}^w$ ,  $m_{ij}^a$ , and  $m_{ij}^n$  indicate whether pair  $(i, j)$  has valid same-work, same-agenda, and agenda-non-identity supervision, respectively. The model minimizes the masked objective

$$\begin{aligned}
Z &= \sum_{(i,j)} (m_{ij}^w + m_{ij}^a + m_{ij}^n), \\
\mathcal{L} &= \frac{1}{Z} \sum_{(i,j)} [\lambda_w m_{ij}^w \text{CE}(y_{ij}^I, p_{\text{work}}) \\
&\quad + \lambda_a m_{ij}^a \text{CE}(y_{ij}^A, p_{\text{agenda}}) + \lambda_n m_{ij}^n \text{CE}(y_{ij}^{ANI}, p_{\text{ani}})],
\end{aligned}$$

where CE is binary cross-entropy and  $\lambda_w, \lambda_a, \lambda_n$  control the contribution of each supervision channel. Missing labels therefore do not create negative

examples for unrelated heads. A mini-batch with  $Z = 0$  is skipped before the optimizer update, and each reported relation head requires positive mask coverage in the declared training split. DeepMatcher pairs supervise same-work identity, whereas OpenAlex/OpenCitations silver hard negatives supervise agenda-non-identity risk without being promoted to human gold identity labels. The false-merge risk score is not directly supervised; it is derived after the relation heads are estimated.

### *5.6. Training and Inference Trace*

The implementation keeps training, threshold fixation, inference, and metric export in one auditable trace. This trace is necessary because IAD-Risk is a selective decision system: the same row-level decisions that produce same-work F1 also determine FMR, HNFMR, automatic merge coverage, block rate, and defer rate. Table 7 gives the execution contract that reviewers should use when checking whether a result artifact matches the method description.

Table 7: Training and inference trace for IAD-Risk. Each phase states the required operation and the artifact or invariant that makes the phase auditable.

| Phase              | Required operation                                                                                       | Auditable output or invariant                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Schema loading     | Load pair records with relation labels, label strength, provenance, split, and hard-negative fields.     | Pair IDs and split fields remain unchanged from input to prediction output.                      |
| Supervised fitting | Optimize the masked objective only where the relevant supervision mask is valid.                         | Gold, proxy, and silver labels are not silently converted into labels for unrelated heads.       |
| Threshold fixation | Select or record $\tau_w$ , $\tau_a$ , and $\tau_r$ before held-out reporting.                           | Threshold source, value, selection split, and selection metric are written to the threshold log. |
| Pair scoring       | Predict $p_{\text{work}}$ , $p_{\text{agenda}}$ , and $p_{\text{ani}}$ , then derive $p_{\text{risk}}$ . | Prediction rows expose relation scores and the derived risk value for each pair.                 |
| Decision emission  | Apply cannot-link evidence and the risk-aware merge gate to emit merge, block, or defer.                 | Decision rows record threshold values, threshold source, and cannot-link status.                 |
| Metric export      | Count same-work F1, FMR, HNFMR, coverage, block rate, and defer rate from the same decision file.        | Metric summaries and checksums bind denominators to the prediction artifact.                     |

### 5.7. Risk-Aware Merge Decision

The merge policy is:

$$\hat{m}_{ij} = 1 \quad \text{iff} \quad p_{\text{work}} \geq \tau_w, \quad p_{\text{ani}} < \tau_a, \quad p_{\text{risk}} < \tau_r, \quad \text{cannot\_link}_{ij} = 0.$$

This rule is deliberately selective. It can reject or defer high-risk pairs instead of forcing every semantically related pair into a binary match decision. The same decision interface also supports constrained union-find clustering, where cannot-link evidence can block documented conflicts when coverage is available. Operationally, the three outputs have different meanings. A pair is merged only when all gate conditions pass. A pair is blocked when explicit cannot-link evidence, high agenda-non-identity evidence, or high derived risk

makes an automatic merge unsafe under the fixed operating point. A pair is deferred when identity evidence is insufficient for automatic merging but the available evidence does not justify a hard block. This distinction lets the method control false merges without claiming that every unresolved pair has been fully adjudicated.

### 5.8. Scoring and Merge Algorithm

The executable method follows a fixed scoring and merge order. For each candidate pair, IAD-Risk first builds identity, agenda, ANI, and audit fields from the pair schema. It then predicts the three relation heads, computes the derived risk score, applies cannot-link evidence, and emits one of three decisions: merge, block, or defer. Table 8 states the algorithmic contract used by the implementation and the artifact audit.

Table 8: Scoring and merge algorithm for IAD-Risk. The table gives the execution order that connects pair inputs, relation-head outputs, risk gating, and metric-counting artifacts.

| Step | Operation                                                        | Required inputs or outputs                                                           | Audit role                                                          |
|------|------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1    | Load a candidate pair and schema metadata.                       | Pair ID, document IDs, label strength, provenance, split, and hard-negative level.   | Fixes row identity before scoring.                                  |
| 2    | Build feature groups without using audit metadata as predictors. | Identity, agenda, ANI, and cannot-link feature groups.                               | Prevents provenance or split leakage.                               |
| 3    | Predict relation heads.                                          | $p_{\text{work}}$ , $p_{\text{agenda}}$ , and $p_{\text{ani}}$ .                     | Records interpretable relation evidence.                            |
| 4    | Compute derived false-merge risk.                                | $p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}$ . | Makes the blocking score reproducible.                              |
| 5    | Apply the merge gate and cannot-link evidence.                   | $\tau_w$ , $\tau_a$ , $\tau_r$ , cannot-link flag, and threshold source.             | Emits merge, block, or defer under a fixed operating point.         |
| 6    | Write prediction and metric audit fields.                        | Decision, row scope, denominators, thresholds, and checksum-bound artifact row.      | Supports same-work F1, FMR, HNFMR, coverage, and defer-rate audits. |

### *5.9. Operational Complexity and Net Benefit*

IAD-Risk is appropriate when false merges are more costly than additional review. The additional cost comes from three relation heads and explicit threshold records, rather than from a new end-to-end clustering system. In an operational scholarly index, this cost is justified when a false merge can contaminate citation counts, literature-review evidence, or recommendation signals. Net benefit also requires workload evidence: automatic merge coverage must be large enough to reduce reviewer work after accounting for blocked and deferred pairs, and the deferral budget and manual-review capacity must be recorded before deployment. Low FMR or HNFMR alone is insufficient for productivity or cost-saving claims. Thresholds are shared operating-point parameters selected on validation evidence, not tuned per pair. Without workload evidence, IAD-Risk should be read as a conservative safety filter, not a universal replacement for simple deduplication pipelines. The net benefit is strongest in high-stakes scholarly indexes, whereas low-risk bulk cleanup tasks may prefer simpler rules. The full operational net-benefit matrix is reported in the supplementary material.

### *5.10. Version and Identifier Boundary*

Version and identifier evidence is treated as merge-policy input rather than as a post-hoc cleanup rule. DOI, arXiv, and OpenAlex identifiers provide strong identity cues when they agree, but publication-lineage evidence is still needed when records describe preprint, conference, and journal versions of related work. In this policy, identifier agreement supports merge eligibility, identifier conflict creates cannot-link or defer evidence, and not every related version is automatically the same work. IAD-Risk therefore separates identifier agreement, identifier conflict, and uncertain version boundaries before applying cluster-level merging. The version policy must be declared before cluster-level merging, and manual adjudication is required for ambiguous version boundaries. The full version and identifier boundary table is reported in the supplementary material.

### *5.11. Failure-Control Rationale*

IAD-Risk is designed as a failure-control framework rather than as another similarity scorer. Table 9 summarizes how each design choice addresses a specific false-merge pathway. The table also states the remaining boundary for each control, because a safety mechanism should make its unsupported cases explicit.



Table 9: Failure-control rationale of IAD-Risk. Each component targets a concrete way in which agenda-level similarity can be mistaken for work identity.

| Failure pathway                                                                                               | Design response                                                                                                                                            | Remaining boundary                                                                                                                |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Topically close papers receive high semantic similarity. Silver metadata is treated as if it were human gold. | Separate agenda evidence from identity evidence before the merge gate. Preserve label strength and provenance, and mask supervision by valid label source. | Requires the benchmark to include agenda-level hard negatives. Silver labels support stress testing, not exhaustive truth claims. |
| Pairwise errors contaminate clusters through transitivity.                                                    | Apply risk gating before constrained union-find and respect cannot-link evidence.                                                                          | Cluster-level guarantees require complete cannot-link coverage.                                                                   |
| Thresholds turn a classifier into an unsafe automatic merger.                                                 | Expose work, ANI, and risk thresholds and evaluate held-out merge decisions.                                                                               | Threshold transfer should be rechecked under new source distributions.                                                            |
| Proxy labels are over-interpreted.                                                                            | Keep proxy rows separate unless the label source supports the relation being evaluated.                                                                    | Proxy rows remain non-human evidence even when reproducible.                                                                      |

### 5.12. Design Alternatives and Rejected Shortcuts

Several simpler designs are plausible but unsafe for the claim made in this paper. Table 10 explains why IAD-Risk uses explicit identity, agenda, and agenda-non-identity signals instead of treating scholarly deduplication as ordinary semantic thresholding or a single pair-classification problem. The purpose of this comparison is not to dismiss these alternatives; some are strong baselines in the reported evidence. The point is that they do not expose the relation semantics needed to audit agenda-driven false merges.

Table 10: Design alternatives considered during method design. Each rejected shortcut lacks an explicit mechanism for separating agenda relatedness from work identity or for auditing false-merge risk.

| Alternative                                 | Why it is insufficient for this failure mode                                                                                     | IAD-Risk design response                                                                         | Evidence boundary                                                     |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Tune a representation-similarity threshold. | High scientific relatedness can still connect different works in the same task, benchmark, or citation community.                | Use representation similarity as candidate and agenda evidence, not direct merge evidence.       | Supports false-merge-risk motivation, not a full retrieval benchmark. |
| Use one supervised pair classifier.         | A strong classifier can reduce false merges but does not expose whether a positive score comes from identity or agenda evidence. | Predict same-work, same-agenda, and agenda-non-identity signals separately before gating.        | RoBERTa remains a strong baseline; broad superiority is not claimed.  |
| Use provenance as a model feature.          | Source or split identifiers can create leakage and make held-out interpretation fragile.                                         | Keep provenance as audit metadata and apply provenance-aware supervision masking.                | Source-heldout claims require stronger source coverage.               |
| Always force a binary merge decision.       | Some pairs have related evidence but insufficient identity evidence, making automatic merging risky.                             | Emit merge, block, or defer through the risk-aware gate.                                         | Defer behavior needs deployment-specific review policies.             |
| Select thresholds after test results.       | Post-hoc threshold choice can hide false-merge risk and inflate held-out conclusions.                                            | Fix operating points before final reporting and require threshold artifacts for stronger claims. | Threshold stability needs a released grid and checksums.              |

### 5.13. Implementation Details

The reported transformer variants use frozen scientific document encoders rather than end-to-end fine-tuning. For each candidate pair, the implemen-

tation builds three feature groups: identity features combine transformer distances with identifiers, author overlap, venue, and year signals; agenda features combine semantic similarity with topic and citation overlap; agenda-non-identity features add conflict indicators such as different identifiers and year conflict. The three relation heads are trained on the training split, and the merge policy exposes  $\tau_w$ ,  $\tau_a$ , and  $\tau_r$  as configurable thresholds. Unless a validation protocol overrides them, the implementation uses 0.5 for all three thresholds.

#### *5.14. Feature and Head Specification*

The implementation uses label provenance for supervision control, not as a predictive shortcut. Pair metadata such as label source, label strength, provenance, split, and hard-negative level are copied to prediction outputs for auditing, but they are excluded from the identity, agenda, and risk feature groups used by the relation heads. Table 11 gives the implementation-level contract used by the reported transformer variants.

Table 11: Feature and head specification for IAD-Risk transformer variants. Metadata fields are retained for auditing but are not used as model features.

| Head or stage  | Main input fields                                                                                                                                           | Supervision or calculation                                                                    | Output role                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Identity head  | Transformer distances, title similarity, author overlap, first-author match, venue similarity, year match, DOI/arXiv/OpenAlex identifier agreement.         | Same-work labels from valid identity evidence, with provenance-aware masking.                 | Estimates $p_{\text{work}}$ as positive merge evidence.                           |
| Agenda head    | Transformer cosine, title and venue similarity, OpenAlex-style topic overlap, reference Jaccard similarity, year-distance score.                            | Same-agenda proxy or silver evidence when the source supports agenda relatedness.             | Estimates $p_{\text{agenda}}$ as relatedness evidence, not direct merge evidence. |
| ANI risk head  | Transformer distances, title and author signals, topic overlap, reference overlap, year conflict, identifier agreement, and different-identifier conflicts. | Agenda-non-identity hard negatives from valid hard-negative sources.                          | Estimates $p_{\text{ani}}$ as direct blocking evidence.                           |
| Risk gate      | $p_{\text{work}}$ , $p_{\text{agenda}}$ , $p_{\text{ani}}$ , $\tau_w$ , $\tau_a$ , $\tau_r$ , and cannot-link flags.                                        | Computes $p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}$ . | Emits merge, block, or defer decisions.                                           |
| Audit metadata | Pair ID, source pair ID, label strength, label source, provenance, split, and hard-negative level.                                                          | Copied from pair records and result artifacts.                                                | Supports artifact review; it is not a training feature.                           |

## 6. Experiments

### 6.1. Dataset Construction and Evidence Strength

Open-v2 is built by converting public scholarly metadata into a unified pair schema. DeepMatcher/py\_entitymatching records provide the same-work gold layer, while OpenAlex topics and OpenCitations-style citation signals provide agenda-level silver hard negatives. Each pair keeps its label strength, provenance, split, and hard-negative flag so that a gold identity

result is never averaged with a silver stress-test result without being identified as such.

The benchmark therefore has two roles. The gold layer measures ordinary duplicate matching ability, and the silver hard-negative layer measures whether a system over-merges topically close but distinct papers. This separation is important for interpreting the results: a representation baseline can look useful for scientific retrieval while still being unsafe for automatic work identity merging.

### *6.2. Split and Leakage Controls*

The evaluation protocol treats split assignment as part of the evidence contract rather than as a preprocessing detail. Each pair carries a split field, label strength, label source, relation label, provenance, and hard-negative level. Training uses only the declared training split, threshold selection uses validation evidence when available, and held-out metrics are reported only after the operating point has been fixed. Metadata fields that identify source, provenance, or split are retained for auditing but are excluded from model features.

The full split and leakage controls table is reported in the supplementary material. The control set includes the train/dev/test split field, an Unordered pair leakage guard, a Label-stratum coverage audit, a Source-heldout readiness audit, and a Topic-heldout readiness audit. These controls define what the current evidence supports and what stronger follow-up evidence would require before held-out, source-generalization, or topic-stability claims are made.

These controls reduce two common validity risks. First, the model cannot gain accuracy from explicit provenance or split identifiers because those fields are audit metadata rather than predictive features. Second, a held-out result is interpreted only within the split and coverage regime that generated it. If a split-readiness audit blocks source-heldout or topic-heldout claims, the manuscript keeps the conclusion at the level of random-split or mechanism evidence instead of presenting it as broad generalization.

### *6.3. Experimental Questions*

The experiments ask four questions. RQ1 tests whether IAD-Risk preserves same-work matching performance on gold labels. RQ2 tests whether it reduces false merges on agenda-level hard negatives. RQ3 examines whether the observed behavior is consistent with the proposed risk mechanism. RQ4

tests whether results remain interpretable under gold, proxy, and silver label strata.

Table 12: Evaluation protocol. Each question is tied to a label stratum and a metric so that evidence strength is not mixed across sources.

| RQ  | Evidence layer           | Metric        | Interpretation              |
|-----|--------------------------|---------------|-----------------------------|
| RQ1 | Gold identity pairs      | Same-work F1  | Duplicate matching ability  |
| RQ2 | Silver hard negatives    | HNFM          | False-merge safety          |
| RQ3 | Mechanism analysis       | FMR and HNFM  | Role of risk signals        |
| RQ4 | Gold/proxy/silver strata | Split metrics | Provenance-aware robustness |

These questions are ordered by the strength of evidence that the current package can support. RQ1 and RQ2 are the primary quantitative checks because they map directly to gold identity rows and silver hard-negative rows. RQ3 is a mechanism-consistency check, not a complete causal ablation claim. RQ4 is an interpretation check over label strata, not a claim that source-heldout or topic-heldout generalization has been fully established. This ordering keeps the result section focused on bounded false-merge evidence rather than on an unsupported same-scope method ranking.

#### 6.4. Evaluation Metrics

Same-work F1 is computed on pairs whose identity labels come from the gold layer. Ordinary false-merge rate (FMR) is the fraction of non-identity pairs that the system marks for automatic merging. Hard-negative false-merge rate (HNFM) is the same error measured only on agenda-level hard negatives, where topic or citation evidence is high but identity evidence is negative. Reporting both FMR and HNFM separates general matching errors from the specific identity-agenda confusion that this paper studies.

#### 6.5. Decision-to-Metric Mapping

The reported metrics treat the selective merge decision as a pair-level prediction. In this mapping, automatic merge is the positive decision, while block and defer are non-merge decisions. Deferred same-work pairs reduce

recall in same-work F1, so a system cannot improve F1 by deferring positive identity pairs without penalty. FMR and HNFMR count only automatic merges among non-identity rows, while coverage and defer rate must be reported separately before operational automation claims are made. The full decision-to-metric mapping table is reported in the supplementary material. Operationally, merge is a positive prediction for same-work F1 and is counted as an FMR/HNFMR error when the label is non-identity; block suppresses false merges but does not resolve a positive same-work pair; and defer protects against unsafe automatic decisions but shifts the case to review and lowers recall when the pair is truly same-work.

#### *6.6. Metric Formula Boundary*

The metric formulas use the selective merge output after thresholding. Let TP denote same-work pairs that are automatically merged, FP denote non-identity pairs that are automatically merged, and FN denote same-work pairs that are blocked or deferred. Same-work F1 is computed as  $2TP/(2TP + FP + FN)$  on the gold identity evaluation rows, where defer and block decisions on true same-work rows enter FN. FMR denominator is all non-identity rows in the evaluated scope and measures unsafe automatic merges rather than manual-review workload, while HNFMR denominator is the agenda-level hard-negative subset and measures identity-agenda false merges under that stratum. Rows excluded by missing labels are not silently added to denominators; they remain outside the corresponding metric until a valid label stratum is available. The full metric-formula boundary table is reported in the supplementary material.

#### *6.7. Threshold Selection and Uncertainty Reporting*

Threshold-sensitive systems are evaluated under a fixed selection protocol rather than by choosing the best test threshold after observing results. Merge thresholds are selected on validation evidence when validation rows are available; otherwise, the default implementation threshold is reported as a fixed operating point. The test split is then used only for final metric reporting. This protocol is necessary because a small threshold change can alter FMR and HNFMR more strongly than it alters same-work F1. The full threshold and uncertainty reporting protocol is reported in the supplementary material. Stronger uncertainty or mechanism claims require prediction files and resampling logs, ablation outputs for no-risk-gate, no-ANI, and single-space

variants, and an artifact audit that binds predictions, configs, logs, checksums, and commit identifiers to each reported table.

#### 6.8. *Statistical Interpretation Boundary*

The reported Open-v2 numbers are point estimates for a fixed evidence snapshot, not statistical superiority estimates. Confidence intervals, significance tests, and model-ranking statements are intentionally withheld unless the released artifact package contains the exact prediction files, resampling logs, random seeds, and checksums used to compute them. In particular, an HNFMR value of 0.000 means that no hard-negative false merge was observed under the reported pair scope and operating point; it should not be read as proof of zero risk under all source distributions. Confidence-interval claims require `bootstrap_intervals` generated from the same prediction files. Pre-defined tests, multiplicity handling, input artifacts, and reproducible analysis logs are required before statistical-significance wording. Same-scope predictions, interval estimates, ablations, and manual-validation slice are required before stronger model-ranking claims. The full statistical interpretation table is reported in the supplementary material.

#### 6.9. *Operating Point Disclosure*

The main result table reports fixed operating points rather than post-hoc best test thresholds. The full operating-point disclosure table is reported in the supplementary material. The main text keeps the operating-point rule as prose: Representation cosine baselines use cosine score thresholds and require a score file, metric summary, and threshold entry; the RoBERTa pair classifier uses a pair match probability threshold and requires a prediction file, metric summary, and model log; and IAD-Risk transformer variants use  $p_{\text{work}}$ ,  $p_{\text{ani}}$ , and  $p_{\text{risk}}$  under the risk gate, with default  $\tau_w = \tau_a = \tau_r = 0.5$  unless the artifact config records an override. Exact threshold values, selected validation sources, command logs, prediction file, model JSON, thresholds, and checksums must be verified from the released artifact manifest before using the table for numerical reproduction.

#### 6.10. *Selective Decision Coverage Boundary*

IAD-Risk is a selective decision system, so false-merge metrics should be interpreted together with coverage and review-load evidence. A low FMR or HNFMR is not by itself a throughput claim if the operating point blocks or defers many pairs. A result with low FMR or HNFMR but high deferral is a



conservative triage result rather than evidence of reduced review cost. The full selective-decision coverage boundary table is reported in the supplementary material. The main text keeps the operational boundary as a prose rule: automatic merge coverage, block rate, defer rate, review load, and capacity-normalized review load must be computed from the same prediction files and compared with a predeclared manual-review capacity and deferral budget before making operational automation claims. The current manuscript therefore reports false-merge control under fixed operating points, but does not claim throughput reduction and does not claim all-pair automatic resolution.

#### *6.11. Pair-to-Cluster Evidence Boundary*

The reported pair-level metrics do not by themselves prove cluster-level deduplication quality. Pair decisions can propagate through transitive merge propagation when a deduplication system turns accepted links into connected components. A single unsafe merge can contaminate a larger cluster, while one cannot-link decision can block a transitive path that would otherwise merge distinct works. The full pair-to-cluster evidence boundary table is reported in the supplementary material. The main text keeps the deployment boundary as a prose rule: cluster assignments, cannot-link violations, `cannot_link_audit`, `cluster_metric_summary`, pair-to-cluster trace files, and cluster contamination rate are required before pair-level false-merge control can be interpreted as a cluster-level deployment guarantee. The paper therefore does not claim cluster-level contamination is eliminated.

#### *6.12. Threshold Sensitivity Evidence Status*

Threshold stability is treated as an audit requirement, not as an unsupported robustness claim. The current Open-v2 table reports fixed operating points and therefore supports the mechanism interpretation under those operating points. The full threshold-sensitivity evidence status table is reported in the supplementary material. The main text keeps the threshold boundary as a prose rule: the current manuscript reports fixed operating points, the threshold grid is not reported as primary evidence, and metric stability requires the same prediction files, predefined threshold ranges, per-threshold F1, FMR, HNFMR, pair counts, random seeds, command logs, a manifest, and checksums. The current result therefore supports fixed-threshold false-merge control, not threshold-stable ranking across all operating points.

### 6.13. *Baselines*

The main table reports two scientific representation cosine baselines and one supervised pair-classifier baseline: SciNCL cosine, SPECTER2 adapter cosine, and RoBERTa pair classification. These rows are evaluated through the same pair schema so that same-work F1 and hard-negative false-merge rate can be interpreted without mixing label strengths. Other repository baseline utilities are not used as primary manuscript evidence unless their metric summaries, prediction files, thresholds, and checksums are included in the released artifact package.

### 6.14. *Baseline Inclusion Rationale*

The baseline set is selected by evidence availability and evaluation relevance rather than by convenience. The complete inclusion matrix is reported in the supplementary material so that the main manuscript states only the decision rule. The prose keeps exact identifier matching and title-normalization rules as important sanity checks; their current boundary is: Excluded from primary result table when only utility code or fixture-level checks are available. Traditional entity-resolution systems use the artifact rule: Included as primary evidence only when metric summaries, prediction files, threshold records, and checksums are available. Scientific representation baselines are included because SciNCL cosine and SPECTER2 adapter cosine have reported score summaries under Open-v2, and RoBERTa pair classification is included as a supervised comparator while keeping the result below broad ranking claims.

Omitting a baseline family from Table 14 is not a claim that omitted baselines were outperformed. It means the current manuscript does not yet have the artifact-backed evidence needed to make that row auditable. A same-scope baseline matrix is required before broad ranking claims, especially for exact identifier matching, title-normalization rules, and traditional entity-resolution systems.

### 6.15. *Baseline Fairness Controls*

Baseline comparison is controlled through the pair contract rather than through a broad comparative-ranking claim. All baselines consume the same IAD-Bench pair records for the row scope being reported, and each row uses the same train/dev/test split field when training is required. Threshold-sensitive rows use validation-selected operating points before held-out reporting. All systems report same-work F1, FMR, and HNFMR from the

same metric implementation. Label source, provenance, and split identifiers remain audit fields, not predictive features. A stricter comparative ranking requires same-scope released prediction files, and rows with different available scopes should not be read as a single comparative ranking. The full fairness-control matrix is in the supplementary material.

#### *6.16. Implementation and Reproducibility*

All reported systems are evaluated through the same pair-level schema, split field, and metric implementation. Threshold-sensitive baselines select merge thresholds on validation evidence and are then evaluated on held-out evidence. The main metrics are same-work F1 on gold identity pairs, ordinary false-merge rate, and HNFMR on agenda-level hard negatives. Rows with different pair counts in Table 14 should therefore be read as evidence for the same mechanism under different available result scopes, not as a single comparative ranking.

The repository separates executable code from large data artifacts: raw third-party data remain outside Git, while processing scripts, fixture tests, schema contracts, and artifact-release rules remain version controlled. Reproduction starts from public sources, rebuilds IAD-Bench pair files through the documented CLI, and checks generated outputs with manifests and checksums. This organization is intended to make the main evidence reproducible without redistributing restricted or large source files. The supplementary material states the claim-evidence boundary for each major conclusion so that code-level reproducibility, artifact-level reproducibility, and manual-validation requirements are not conflated.

#### *6.17. Artifact Reproduction Protocol*

Because the repository does not redistribute large third-party data, reproducibility is divided into code-level reproduction and result-level artifact reproduction. Table 13 defines what each level proves. The key principle is that small fixtures verify the processing contracts, while the full reported tables require either independently obtained public source files or a released artifact package with manifests and checksums.

Table 13: Reproduction levels for auditing the repository and the reported evidence.

| Level                    | Required input                                                                                                                              | Audit target                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| L0 code check            | Source tree, package environment, and manuscript scripts                                                                                    | Verify that the CLI, tests, public-release scan, and LaTeX build run in a clean environment.                               |
| L1 fixture rebuild       | Public miniature fixtures under <code>tests/fixtures</code>                                                                                 | Verify that DeepMatcher, SciRepEval-style, OpenAlex-style, and IAD-Bench adapters produce the documented pair schema.      |
| L2 public-source rebuild | Independently obtained public raw files from the cited sources, plus <code>source_input_manifest</code> and <code>processing_run_log</code> | Rebuild derived IAD-Bench files with documented commands, input provenance, run metadata, output summaries, and checksums. |
| L3 result audit          | Released tables, predictions, logs, manifests, and checksums                                                                                | Verify the numerical results reported in the manuscript against immutable artifact files.                                  |

#### 6.18. Claim-Evidence Boundary for Result Interpretation

The result interpretation follows a conservative claim-evidence boundary. Each reported conclusion must be tied to a label stratum, an output artifact, and a statement of what the evidence does not prove. This boundary is especially important because the benchmark combines gold identity pairs with proxy and silver stress-test evidence rather than treating all rows as one homogeneous test set.

The full claim-evidence boundary table is reported in the supplementary material. The main text keeps the boundary as a prose rule: identity-agenda confusion is supported only as a false-merge pathway; IAD-Risk support is bounded to the reported Open-v2 setting; IAD-Bench is a provenance-aware evaluation contract; and repository-level reproduction does not by itself prove full numerical results without public source files or a released artifact package.

#### 6.19. Result Audit Trail

Table 14 should be read together with the claim-evidence boundary stated above and expanded in the supplementary material. Each numerical row

must be traceable to three audit objects: a prediction or score file, a metric summary produced from that file, and a checksum or manifest entry that fixes the artifact version. The same audit record must also contain per-row denominator counts for same-work F1, FMR, and HNFMR, the per-row threshold source, the scope label used in the main table, and the selective-decision coverage fields: automatic merge count, block count, defer count, automatic merge coverage, defer rate, and capacity-normalized review load. Prediction JSONL artifacts must expose system, pair/document IDs, expected labels, label strength, hard-negative level, split identifiers, score or probability fields, threshold source, threshold values where applicable, and merge decisions; threshold logs must expose threshold name, value, selection split, selection metric, selection rule, applied scope, and score field. Public-source rebuild evidence must also include `source_input_manifest` and `processing_run_log`, because these files bind public inputs, commands, code commits, environments, seeds, output paths, and exit status to the derived evaluation artifacts. The Git repository verifies the schema, fixture rebuild, public-release boundary, manuscript build, and submission package, but it deliberately does not store raw third-party data, large model outputs, or full prediction files. Therefore the manuscript treats the Open-v2 table as a supported evidence snapshot and reserves full numerical reproduction for the released L3 artifact package.

The complete row-family artifact crosswalk is reported in the supplementary material. It binds `open_v2_main_results`, `iad_bench_split_summary`, `representation_baseline_scores`, `supervised_baseline_predictions`, `iad_risk_predictions`, `threshold_selection_logs`, `bootstrap_intervals`, `ablation_suite`, `manual_validation_slice`, and `threshold_sensitivity_grid` to the row scopes, operating points, optional stronger claims, and checksum-fixed artifact records used for full numerical audit.

## 6.20. Main Open-v2 Results

Table 14: Open-v2 evidence snapshot. Lower false-merge rate is better. Rows are reported only when supported by current output summaries; the scope type column marks whether a row uses the full available Open-v2 scope or the held-out Open-v2 test scope. The Denom. audit column means that same-work F1, FMR, and HNFMR denominators must be present in the corresponding `open_v2_main_results` artifact row before numerical audit.

| System                  | Scope type    | Pairs Denom. audit | F1 ↑ FMR ↓ HNFMR ↓ |
|-------------------------|---------------|--------------------|--------------------|
| SciNCL cosine           | Full Open-v2  | 10415 Artifact row | 0.054 0.785 0.790  |
| SPECTER2 adapter cosine | Full Open-v2  | 10415 Artifact row | 0.044 0.999 0.999  |
| RoBERTa pair classifier | Full Open-v2  | 10415 Artifact row | 0.825 0.001 0.0001 |
| IAD-Risk (SciNCL)       | Held-out test | 1042 Artifact row  | 0.980 0.001 0.000  |
| IAD-Risk (SPECTER2)     | Held-out test | 1042 Artifact row  | 0.980 0.001 0.000  |

Figure 2 visualizes the same Open-v2 evidence as Table 14. The figure is intended as a mechanism-evidence profile: it makes the hard-negative false-merge exposure of single-space representation baselines visible while keeping the full-scope and held-out test rows separated.



Figure 2: Open-v2 mechanism-evidence profile. The bars visualize Table 14 without changing its interpretation boundary: full Open-v2 rows and held-out test rows remain scope-labeled, and the figure supports the false-merge-control reading rather than a same-scope comparative ranking.

The Open-v2 evidence shows why identity-agenda disentanglement is necessary. Single-space scientific representation baselines preserve topical relatedness, but their merge threshold can produce high false-merge rates on OpenAlex silver hard negatives. IAD-Risk transformer variants keep high same-work F1 on the held-out Open-v2 test split while showing zero observed hard-negative false merges under the reported operating point. RoBERTa pair classification is a strong baseline and narrows the gap substantially; therefore the paper does not make broad superiority claims. Instead, the main result is that relation heads plus an explicit risk score provide a transparent and controllable mechanism for false-merge suppression.

### 6.21. Result Interpretation Guardrails

The full result interpretation guardrails table is reported in the supplementary material. The Open-v2 table should be read before external artifact release and same-scope prediction files are available. The main result table includes Scope type labels to distinguish full available Open-v2 scope rows from held-out Open-v2 test scope rows. Scope labels prevent ranking interpretation when row families are supported by different available prediction scopes. The intended interpretation is mechanism-oriented: representation rows test false-merge exposure, the RoBERTa row is a strong supervised comparator, and IAD-Risk rows test split-held-out risk gating with explicit identity,

agenda, and ANI signals. The table is not a claim of broad method superiority, not a same-scope comparative ranking, and not evidence of threshold stability or zero risk.

#### *6.22. Manual Validation Boundary*

Manual validation is not completed in the current manuscript package. Silver hard negatives are stress-test evidence for identity-agenda false-merge risk, but they are not human-gold non-identity labels. The full manual validation boundary table is reported in the supplementary material. Stronger label-precision claims require a 500–1,000 pair reviewed slice covering hard negatives, high-score false-merge candidates, blocked or deferred pairs, version boundaries, identifier conflicts, and sparse metadata. The protocol requires two independent reviewers who are blind to model scores and merge decisions, plus an adjudication log, agreement report, pair-level notes, and checksums. The paper therefore supports bounded false-merge stress testing under Open-v2, but does not claim complete human validation or that silver rows replace human-gold non-identity labels.

#### *6.23. Scope Compatibility of the Open-v2 Table*

The Open-v2 table is a scope-bounded evidence table, not a single comparative ranking. Full pair-scope representation baselines test whether single-space scientific relatedness can over-merge agenda-level hard negatives across the available Open-v2 pairs. Held-out IAD-Risk rows test whether risk-gated transformer variants preserve same-work performance and suppress hard-negative merges under a split-based model-evaluation protocol. The valid comparison is therefore mechanistic: semantic relatedness alone is unsafe as automatic merge evidence, while explicit risk gating can block hard-negative merges in the reported held-out setting.

The full scope compatibility matrix is reported in the supplementary material. This scope boundary is important for interpretation. The evidence is sufficient for the paper’s conservative claim that identity-agenda disentanglement is a practical false-merge control mechanism. It is not sufficient for a broad ranking claim until all row families are regenerated with the same released prediction scope, threshold-selection logs, checksums, and an independent manual-validation slice. A claim that this stronger comparison has already been completed is outside the current evidence snapshot.



#### 6.24. *Extended Protocols and Boundaries*

Open-v3 and source-heldout protocols are retained as follow-up evaluation paths, not as additional result evidence in the current manuscript package. These protocols are useful because they specify how source-diverse and domain-diverse validation should be audited before broader claims are made. The manuscript therefore treats Open-v2 as the core mechanism demonstration and reserves Open-v3/source-heldout conclusions for a released artifact package that contains matched prediction scopes, threshold logs, checksums, and manual-validation evidence.

### 7. Mechanism and Error Analysis

The central mechanism question is whether false-merge control follows from explicit agenda-non-identity supervision and the derived risk score. The supported Open-v2 comparison gives mechanism-consistent evidence: single-space scientific representations can over-merge agenda-related hard negatives, while IAD-Risk makes agenda-non-identity evidence a direct blocking signal. This does not by itself establish a complete causal ablation. The full mechanism-evidence boundary table is reported in the supplementary material. The main text keeps the mechanism boundary as a prose rule: topical relatedness creates merge risk only as a targeted failure-mode claim; explicit risk gating supports the reported Open-v2 contract and thresholds; component-causality claims require ablations removing risk gating, ANI supervision, and single-space separation; and cluster-level contamination claims require sufficient cannot-link coverage and cluster-level artifact audits.

The full error taxonomy table is reported in the supplementary material. The main text keeps the taxonomy as a prose rule: same task, different contribution pairs, citation-neighborhood neighbors, version or extension boundaries, identifier conflicts, and sparse metadata cases are the main failure modes the framework is designed to expose. This taxonomy is diagnostic rather than a measured error distribution unless a released artifact contains per-category annotations and adjudication logs. The distinction matters because version boundaries, identifier conflicts, and sparse metadata cases require human judgment beyond metadata-derived silver evidence.

The error analysis emphasizes claim discipline. The current evidence supports identity-agenda risk modeling, stratified benchmark construction, and targeted false-merge suppression. It does not support claims that OpenAlex labels are human gold, that the system has broad domain superiority, that

component-level ablations have already closed every mechanism question, or that manual annotation has already been completed.

## 8. Limitations

This study has five limitations. First, OpenAlex/OpenCitations hard negatives are silver labels, not human gold. They are useful for stress testing but must be reported separately from DeepMatcher gold labels. Second, SciRepEval/SciDocs-style proximity is a proxy for agenda relatedness, not proof of identity or non-identity. Third, source-heldout generalization remains mixed across model variants and should be expanded before making broad claims across scholarly domains. Fourth, pair-level metrics do not by themselves establish cluster-level deployment quality; cluster assignments, cannot-link coverage, and cluster contamination rate are required before making cluster-level quality claims. Fifth, an independently reviewed 500–1,000 pair validation slice would provide stronger evidence about label precision and difficult boundary cases; the supplementary material specifies the sampling, blinding, adjudication, and agreement-reporting protocol required before making stronger human-validation claims.

These limitations define the evidence needed to upgrade the manuscript claims. A stronger label-quality claim requires the reviewed manual-validation slice and adjudication records. A stronger robustness claim requires source-heldout and topic-heldout evaluations generated from the same documented pipeline. A stronger ranking claim requires same-scope prediction files, threshold logs, bootstrap intervals, and checksums for every compared row. A stronger deployment claim requires cluster artifacts that connect pair decisions to final components. Until those artifacts are available, the manuscript should be read as bounded evidence for pair-level false-merge control under the Open-v2 evidence snapshot.

## 9. Threats to Validity

Construct validity depends on whether the label strata measure the intended relations. DeepMatcher pairs provide same-work identity evidence, but OpenAlex/OpenCitations hard negatives are silver stress-test pairs rather than manual non-identity gold. Internal validity depends on threshold selection and split separation; therefore threshold-sensitive systems must select thresholds on validation data and report held-out metrics.

External validity is limited by the current source mix: the Open-v2 evidence supports the identity-agenda mechanism, while broader claims require more source-heldout and domain-diverse evaluation. Reproducibility depends on public data availability and stable processing scripts because the repository deliberately avoids redistributing large third-party data files.

The full validity-threats matrix is reported in the supplementary material. The main text keeps the threat model as a prose rule: construct validity is bounded by label strata; internal validity is bounded by threshold and split separation; external validity is bounded by the current source mix; conclusion validity is bounded by the absence of complete causal ablations; reproducibility is bounded by source and artifact availability; and operational validity is bounded by the gap between pair-level decisions and cluster-level deployment. These safeguards do not turn proxy or silver evidence into human-adjudicated truth, and full numeric audit requires L2/L3 artifacts.

## **10. Claim Interpretation Boundary**

The full claim-interpretation boundary table is reported in the supplementary material. The main text keeps the boundary as a prose rule: contribution clarity is tied to the IAD-Bench contract, identity-agenda confusion, and HNFMR as a false-merge safety problem; writing reproducibility is limited to code-level checks, fixture rebuilds, schema validation, and artifact-release preparation; experimental strength is limited to the Open-v2 evidence snapshot; evaluation completeness is limited by artifact-backed ablations, threshold grids, and a manual-validation slice; and method design soundness remains bounded by source-heldout validation, topic-heldout checks, and failure-case analysis before broad deployment claims.

## **11. Conclusion**

IAD-Risk addresses a specific failure mode in scholarly entity matching: the tendency to confuse agenda similarity with work identity. The core idea is to separate identity and agenda evidence before a pair is allowed to become an automatic merge. By adding agenda-non-identity supervision and a false-merge risk gate, the framework turns scholarly deduplication from an unconditional similarity decision into a selective decision process that can block or defer risky pairs.

The Open-v2 evidence supports a conservative conclusion: identity-agenda disentanglement can provide targeted false-merge suppression under the reported pair scope and operating points. IAD-Bench contributes a reproducible benchmark contract by keeping gold identity labels, proxy agenda evidence, and silver hard negatives separate throughout training, evaluation, and claim interpretation. This separation is the main practical lesson of the study: relatedness evidence is useful for retrieval and candidate generation, but it must not be treated as sufficient evidence of work identity.

The manuscript does not claim cluster-level deployment quality without cluster artifacts, nor does it claim broad method ranking across scholarly domains. Additional validation should expand independent manual review, source-heldout evaluation, same-scope baseline prediction releases, and cluster-level artifact audits. These extensions would make it possible to test whether the present pair-level safety mechanism transfers to broader source mixtures and production-style deduplication workflows.

## Data and Code Availability

The source code, benchmark construction scripts, public-release checks, and manuscript build scripts are organized in the project repository. The installable package source is under `src/iad_sieve`; `pyproject.toml` declares the source root and exposes `iad-sieve = iad_sieve.cli:main`, while `python -m iad_sieve.cli --help` verifies command discovery before fixture rebuild or artifact commands are run. After installation, the reproducibility entry points include CLI commands for DeepMatcher conversion, proximity conversion in the SciRepEval style, weak-label construction from OpenAlex and OpenCitations, IAD-Bench assembly, stratification checks, source-bias diagnostics, baseline execution, IAD-Risk training, bootstrap analysis, and manuscript PDF generation.

Raw third-party data are not redistributed in the repository because the original sources have their own access conditions and licenses. The repository instead provides data-processing commands, schema contracts, fixture tests, and artifact-release guidance so that independent readers can reproduce the benchmark construction from public sources and verify generated artifacts through manifests, checksums, command logs, and fixed commit identifiers. An L2 public-source rebuild should record `source_input_manifest` and `processing_run_log` files so that public input provenance, processing commands, code commits, environments, seeds, output paths, exit status,

and derived-file checksums can be audited without placing raw third-party data in Git.

For reviewers or readers who evaluate only the Git repository, the executable checks are installation, command-line help via `python -m iad_sieve.cli --help`, fixture rebuild validation via `python manuscript/scripts/verify_fixture_rebuild.py`, the public-release audit via `python scripts/check_public_release.py`, and the manuscript validation scripts. These checks establish that the adapters, schemas, fixture rebuild path, public-release boundary, and PDF build are executable from the version-controlled source tree. They do not reproduce the Open-v2 numerical table, model predictions, threshold choices, or bootstrap intervals; those result-level checks require either an L2/L3 public-source rebuild or a released external artifact package.

An external artifact package should be treated as result evidence only after it passes both artifact-level and manuscript-package validation. Before copying generated result files into a release scaffold, `python manuscript/scripts/populate_artifact_release.py` with `--artifact-dir /path/to/release`, `--source-dir /path/to/source-artifacts`, and `--preflight-only` checks that the source artifact directory contains the files required by the release manifest without writing release files. This source preflight can detect missing tables, predictions, reports, configurations, or logs early, but it is not itself result evidence. The command `python manuscript/scripts/validate_artifact_release.py --artifact-dir /path/to/release` checks the release manifest, checksums, required result identifiers, row-level schemas, prediction schemas, claim-boundary flags, and raw-data exclusions. The command `python manuscript/scripts/validate_submission_package.py --final-upload --artifact-dir /path/to/release` then checks that the final manuscript package, submission metadata, source-control commit, and artifact manifest refer to the same source commit. If either validation fails, the artifact package should not be used to support the Open-v2 numerical table or any stronger claim.

The full data and code availability boundary table is reported in the supplementary material. The boundary distinguishes L0/L1 code-level reproduction from L2/L3 result-level audit: source code and CLI entry points, small public fixtures, schema contracts, and data-processing commands are version-controlled in Git, while raw third-party source files, full prediction files, model checkpoints, and derived evaluation artifacts remain outside Git

unless released through the external artifact package. Full numerical reproduction requires public-source rebuilds or released artifacts; the Git repository alone verifies the executable contracts, fixture behavior, data-processing path, and manuscript build rather than redistributing raw data or full experiment outputs.

## **Ethics Statement**

This study uses public scholarly metadata and benchmark-style record linkage data. It does not involve human participants, clinical records, private user behavior, or animal experiments. Any future manual validation should be limited to scholarly record relation labels and should avoid collecting sensitive personal information.

## **Competing Interests**

The competing-interest declaration is not finalized in this anonymous preflight manuscript. Before final upload, the listed authors must confirm the competing-interest status and synchronize the final statement with `submission_metadata.yml` and the live submission system.

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